Sentiment Analysis Pipeline using Python

This experiment classifies text as **Positive, Negative, or Neutral** using the **TextBlob** library.

Sentiment Analysis Pipeline

```
Customer Review
       ↓
Tokenization
       ↓
Text Cleaning
       ↓
Feature Extraction
       ↓
Sentiment Analysis Model
        ↓
 Positive / Negative / Neutral
```

Example

```
Input:
"The movie was excellent and amazing!"

Output:
Positive Sentiment
```

Step 1: Install Required Library

Open Command Prompt / Terminal:

```
pip install textblob
```

Download additional corpora:

```
python -m textblob.download_corpora
```

Step 2: Create the Program

Save the following code as:

```
sentiment_analysis.py
```

```
from textblob import TextBlob

print("Sentiment Analysis System")
print("-" * 30)

while True:

    text = input("\nEnter a review (or type exit): ")

    if text.lower() == "exit":
        break

    analysis = TextBlob(text)

    polarity = analysis.sentiment.polarity

    if polarity > 0:
        sentiment = "Positive 😃"

    elif polarity < 0:
        sentiment = "Negative 😞"

    else:
        sentiment = "Neutral 😐"

    print("Polarity Score:", polarity)
    print("Sentiment:", sentiment)
```

Step 3: Run the Program

Open Terminal in the folder containing the file:

```
python sentiment_analysis.py
```

Sample Execution

Example 1

```
Enter a review:
The movie was fantastic and enjoyable

Polarity Score: 0.6

Sentiment:
Positive 😀
```

Example 2

```
Enter a review:
The service was terrible

Polarity Score: -1.0

Sentiment:
Negative 😞
```

Example 3

```
Enter a review:
The book is on the table

Polarity Score: 0.0

Sentiment:
Neutral 😐
```

How It Works

Input

```
The movie was fantastic
```

TextBlob Analysis

```
analysis = TextBlob(text)
```

Calculates:

```
analysis.sentiment.polarity
```

Range:

Polarity Score	Meaning
+1.0	Very Positive
0	Neutral
-1.0	Very Negative

Lab Output Table

Review	Polarity	Sentiment
The movie was amazing	0.6	Positive
The service was awful	-1.0	Negative
The book is on the table	0.0	Neutral
I love NLP	0.5	Positive
I hate waiting	-0.8	Negative

Viva Questions

1. What is Sentiment Analysis?

Sentiment Analysis is the process of identifying emotions, opinions, or attitudes expressed in text.

2. What are the common sentiment classes?

```
Positive
Negative
Neutral
```

3. What library is used in this experiment?

```
TextBlob
```

4. What is polarity?

Polarity is a numerical score representing the sentiment of a sentence.

```
+1 → Positive
```

```
  0 → Neutral
 -1 → Negative
```

5. Give one real-world application of Sentiment Analysis.

- Product review analysis
- Social media monitoring
- Customer feedback analysis
- Movie review classification
- Brand reputation management

Expected Result

The program successfully analyzes user-entered text and classifies it as:

✅ Positive
✅ Negative
✅ Neutral

based on the polarity score computed by **TextBlob**. This demonstrates a simple **Sentiment Analysis Pipeline**, one of the most common applications of **Natural Language Processing (NLP)**.